



**LACHOO MEMORIAL COLLEGE OF SCIENCE AND TECHNOLOGY**

**UNDEGRADUATE DEGREE PROGRAMME**

**BACHELOR OF COMPUTER APPLICATIONS**

**Learning Outcomes Based Curriculum Framework**

**Choice Based Flexible Credit System**

**ACADEMIC YEAR 2026-27**

### 1. Department Vision Statement

|          |                                                                                                                    |
|----------|--------------------------------------------------------------------------------------------------------------------|
| Stmt - 1 | Creating the most conducive environment for imparting quality education in Computer Applications                   |
| Stmt - 2 | Contributing effectively to produce globally competent quality professionals in the field of Computer Applications |
| Stmt - 3 | Contributing towards preparing young minds to serve community                                                      |

### 2. Department Mission Statement

|          |                                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------|
| Stmt - 1 | Impart student's essential knowledge and skills required for a successful career in Computer Applications                                |
| Stmt - 2 | Instil confidence in the students to take up new challenges by grooming them appropriately                                               |
| Stmt - 3 | Inculcate in the students a sense of commitment to professional ethics, moral values with emphasis on team work and leadership qualities |
| Stmt - 4 | Instil the students with a clear awareness of environmental issues and their relevance to their profession                               |
| Stmt - 5 | Impress upon the students the impact of their work on the nation's economic and social progress                                          |

### 3. Program Education Objectives (PEO)

|         |                                                                                                                                   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|
| PEO - 1 | Offer the students those skill sets and domain knowledge based on needs of Computer Applications and dynamic business environment |
| PEO - 2 | Provide the students with the capabilities in the areas of analysis, design, development and testing                              |
| PEO - 3 | Kindle the minds of students to take up research and development in Computer Applications with missionary zeal                    |
| PEO - 4 | Train the students to become effective communicators in professional as well as general aspects of life                           |
| PEO - 5 | Prepare the students into balanced individuals who are keen to leave a mark by excelling in their profession                      |

### 4. Program Specific Outcomes (PSO)

|         |                                                                                                                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PSO - 1 | Graduates will acquire a comprehensive knowledge and sound understanding of fundamentals of Computer Applications.                                                                                       |
| PSO - 2 | Graduates will develop practical, analytical and programming skills.                                                                                                                                     |
| PSO - 3 | Graduates will be prepared to acquire a range of general skills, to solve problems, to evaluate information, to develop software tools, to communicate with society effectively and learn independently. |

### 5. Consistency of PEO's with Mission of the Department

|                                                                   | Mission Stmt. - 1 | Mission Stmt. - 2 | Mission Stmt. - 3 | Mission Stmt. - 4 | Mission Stmt. - 5 |
|-------------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| PEO - 1                                                           | H                 | H                 | M                 | H                 | M                 |
| PEO - 2                                                           | H                 | M                 | H                 | H                 | H                 |
| PEO - 3                                                           | M                 | H                 | M                 | H                 | H                 |
| PEO - 4                                                           | H                 | H                 | H                 | L                 | M                 |
| PEO - 5                                                           | L                 | H                 | M                 | H                 | H                 |
| H – High Correlation, M – Medium Correlation, L – Low Correlation |                   |                   |                   |                   |                   |

### 6. Consistency of PEO's with Program Learning Outcomes (PLO)

|                                                                   | Program Learning Outcomes (PLO) |                         |                              |                      |                          |                              |                     |                         |                      |                        |                      |                   |            |                        |                    |
|-------------------------------------------------------------------|---------------------------------|-------------------------|------------------------------|----------------------|--------------------------|------------------------------|---------------------|-------------------------|----------------------|------------------------|----------------------|-------------------|------------|------------------------|--------------------|
|                                                                   | 1                               | 2                       | 3                            | 4                    | 5                        | 6                            | 7                   | 8                       | 9                    | 10                     | 11                   | 12                | 13         | 14                     | 15                 |
|                                                                   | Fundamental Knowledge           | Application of Concepts | Link with Related Discipline | Procedural Knowledge | Skills in Specialization | Ability to Utilize Knowledge | Skills in Modelling | Analyse, Interpret Data | Investigative Skills | Problem Solving Skills | Communication Skills | Analytical Skills | ICT Skills | Professional Behaviour | Life Long Learning |
| PEO - 1                                                           | H                               | H                       | H                            | H                    | H                        | L                            | M                   | L                       | M                    | M                      | H                    | H                 | M          | H                      | H                  |
| PEO - 2                                                           | H                               | H                       | H                            | H                    | H                        | L                            | M                   | L                       | M                    | H                      | M                    | M                 | H          | H                      | M                  |
| PEO - 3                                                           | H                               | H                       | H                            | H                    | H                        | M                            | H                   | M                       | M                    | M                      | H                    | H                 | H          | M                      | M                  |
| PEO - 4                                                           | H                               | M                       | M                            | H                    | H                        | H                            | M                   | H                       | H                    | H                      | H                    | L                 | M          | M                      | H                  |
| PEO - 5                                                           | M                               | M                       | H                            | H                    | M                        | H                            | M                   | H                       | H                    | H                      | M                    | M                 | H          | M                      | M                  |
| H – High Correlation, M – Medium Correlation, L – Low Correlation |                                 |                         |                              |                      |                          |                              |                     |                         |                      |                        |                      |                   |            |                        |                    |

## 7. Programme Outcomes

| S. No. | Programme Outcomes                                                                                                                                               |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PO1    | Apply the knowledge and concepts of computer science in day today life.                                                                                          |
| PO2    | Identify, analyze, and formulate computing problems using computer science and application domain.                                                               |
| PO3    | Apply appropriate technologies and tools with an understanding of limitations.                                                                                   |
| PO4    | Recognize the various issues related to society, environment and health and understand the responsibilities to contribute in providing the solutions through IT. |
| PO5    | Assess theoretical foundations of different branches of Computer Science so that students can pursue for higher studies                                          |
| PO6    | Accumulate professional and ethical responsibility.                                                                                                              |
| PO7    | Adapt effectively as an individual, and as a member or leader in diverse/multidisciplinary teams.                                                                |
| PO8    | Pick up good communication and presentation skills                                                                                                               |
| PO9    | Identify the need for and develop the ability to engage in continuous learning as a computing professional.                                                      |

## 8. Programme Structure- B.C. A Computer Applications

|                                                                                          |
|------------------------------------------------------------------------------------------|
| DISCIPLINE SPECIFIC CORE COURSES : CC, MULTI DISCIPLINARY COURSE : MDC                   |
| DISCIPLINE SPECIFIC ELECTIVE COURSE: EC, ABILITY ENHANCEMENT COURSE : AEC                |
| SKILL ENHANCEMENT COURSE : SEC, VALUE ADDED COURSES : VASPL, PROJECT/INTERNSHIP: PRJ/INT |

| SEMESTER                                                                                                            | CC | MDC | EC | AEC | SEC | VASPL | PRJ/ INT | TOTAL CREDITS |
|---------------------------------------------------------------------------------------------------------------------|----|-----|----|-----|-----|-------|----------|---------------|
| I                                                                                                                   | 15 | 4   |    | 4   |     |       |          | 23            |
| II                                                                                                                  | 15 | 4   |    | 4   |     | 3     |          | 25            |
| For exit with certificate in BCA minimum 52 credits (Additional 4 credits of Internship) Total Credits = 48+ 4 = 52 |    |     |    |     |     |       |          |               |

## SEMESTER I

| Course Type          | Course Code | Course Title                           | Teaching Hours Per Week |   |   | Total Credits | CA Marks           | ESE Marks | Total Marks |
|----------------------|-------------|----------------------------------------|-------------------------|---|---|---------------|--------------------|-----------|-------------|
|                      |             |                                        | L                       | T | P |               |                    |           |             |
| DCC                  | CCBCA111    | Programming in C                       | 3                       | 1 | - | 4             | 30                 | 70        | 100         |
| DCC                  | CCBCA112    | Web Design                             | 3                       | - | - | 3             | 30                 | 70        | 100         |
| DCC                  | CCBCA113    | Computer Organization and Architecture | 3                       | 1 | - | 4             | 30                 | 70        | 100         |
| MDC                  | MDBCA111    | Mathematics I                          | 3                       | 1 | - | 4             | 30                 | 70        | 100         |
| AECC                 | AEC         | Hindi / English                        | 4                       | - | - | 4             | 30                 | 70        | 100         |
| DCC                  | CCBCA121    | Programming in C Lab                   | -                       | - | 4 | 2             | 30                 | 70        | 100         |
| DCC                  | CCBCA122    | Web Design Lab                         | -                       | - | 4 | 2             | 30                 | 70        | 100         |
| <b>TOTAL CREDITS</b> |             |                                        |                         |   |   | <b>23</b>     | <b>TOTAL MARKS</b> |           | <b>700</b>  |

## BCA I SEMESTER

|                |                  |           |              |       |
|----------------|------------------|-----------|--------------|-------|
| Semester       | BCA I Sem        |           |              |       |
| Course Code    | CCBCA111         |           | Credits      | 4     |
| Course Name    | PROGRAMMING IN C |           | Pds Per Week | 3     |
| Course Type    | Theory           |           |              |       |
| MID TERM MARKS | 30               | ESE MARKS | 70           | TOTAL |
|                |                  |           |              | 100   |

| S.No. | Course Outcomes                                                                                                                                                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1   | Identify and explain the structure and core components of a C program, including data types, operators, expressions, and the compilation process, to develop basic working programs.                                                      |
| CO2   | Implement programs using control structures (decision-making and loops) and modularize code using functions, including parameter passing methods and recursion.                                                                           |
| CO3   | Design and apply arrays, pointers, and structures for complex data representation. Apply arrays, strings, and pointers to manage collections of data and manipulate memory addresses, utilizing related functions and pointer arithmetic. |
| CO4   | Utilize advanced C features such as preprocessor directives, macros, command-line arguments, and storage classes to write efficient and modular code.                                                                                     |
| CO5   | Develop programs using structures and unions to model complex data and implement file operations for persistent data storage and retrieval.                                                                                               |

| UNIT | PROGRAMMING IN C                                                                                                                                                                                                                                                                                                                                                                           | HRS |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1    | Introduction to C Programming: History and features of C, structure of a C program, creating and executing a C program, compilation process in C, C character set and tokens, data types, declaration and initialization of variables, symbolic constants, formatted and unformatted I/O functions, C operators and expressions, evaluation of arithmetic expressions, and type conversion | 6   |
| 2    | Control Structures and Functions: Decision-making statements :if, if-else, if -else ladder , Loops in C: for, while, do-while , switch case, goto, break and continue statements, Functions : Definition and Declaration, built-in and user-defined functions, nesting of functions, call by value, call by reference, Concept of recursion.                                               | 6   |
| 3    | Arrays, Strings, and Pointers: An Introduction to arrays, one and two dimensional, Storage mechanism ,array declaration and initialization, String Handling :Concept , string functions in C: strlen(), strcpy(), strcat(), strcmp(), String Manipulation, Pointers, pointers and arrays,Accessing one and two dimensional array elements using pointers , pointer arithmetic              | 6   |
| 4    | Advanced Concepts in C: Command-line arguments and their use, argc and argv , macros and preprocessor directives,header files , storage classes :Concepts , Auto, Register , Static and Extern scope and lifetime of variables.                                                                                                                                                            | 6   |
| 5    | Structures, Unions, and File Handling: structures and unions : use ,declarations , comparison of structure and union, array of structure ,array of union, File handling : file input/output functions ,file modes, creating and reading text files,                                                                                                                                        | 6   |

### Suggested Readings:

1. Programming in ANSI C, Balagurusamy, McGraw-Hill
2. Let Us C, Yashawant P. Kanetkar, BPB Publications
3. Programming in C, V. Rajaraman, PHI – EEE
4. C: The Complete Reference, Herbert Schildt
5. The C Programming Language, Kernighan & Ritchie, PHI

|                |                      |           |    |              |     |
|----------------|----------------------|-----------|----|--------------|-----|
| Semester       | BCA I Sem            |           |    |              |     |
| Course Code    | CCBCA121             |           |    | Credits      | 2   |
| Course Name    | PROGRAMMING IN C LAB |           |    | Pds Per Week | 4   |
| Course Type    | Practical            |           |    |              |     |
| MID TERM MARKS | 30                   | ESE MARKS | 70 | TOTAL        | 100 |

**Exercises based on:** Lab Setup and Environment Introduction, Writing and Running Your First C Program, Variables, Data Types, and Basic Input/output Operations, Conditional Statements (if-else), Switch Case Statements, Looping Statements (while, do-while, for), Function Definition and Declaration, Passing Arguments to Functions, Returning Values from Functions, Single-Dimensional Arrays, Multi-Dimensional Arrays, String Manipulation Functions, Introduction to Pointers, Pointer Arithmetic, Pointers and Arrays, Defining and Using Structures, Accessing Structure Members, Introduction to Unions, Basic File Operations (Opening, Reading, Writing, Closing), Command Line Arguments.

|                |            |           |              |           |
|----------------|------------|-----------|--------------|-----------|
| Semester       | BCA I Sem  |           |              |           |
| Course Code    | CCBCA112   |           | Credits      | 3         |
| Course Name    | WEB DESIGN |           | Pds Per Week | 3         |
| Course Type    | Theory     |           |              |           |
| MID-TERM MARKS | 30         | ESE MARKS | 70           | TOTAL 100 |

| S.No. | Course Outcomes                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------|
| CO1   | Identify and describe the key elements and concepts of HTML                                                                     |
| CO2   | Develop hands-on expertise in creating web pages that fulfill specific requirements                                             |
| CO3   | Examine web page structure and content to pinpoint areas for refinement.                                                        |
| CO4   | Appraise the effectiveness of a web page in meeting user needs and provide suggestions for enhancement with CSS                 |
| CO5   | Acquire the expertise to create responsive and interactive web applications using JavaScript to meet real-world needs using DOM |

| UNIT | WEB DESIGN                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | HRS |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1    | Basics of HTML and Page Structure: Introduction of HTML and markup language features, versions of HTML, elements of HTML including empty elements and attributes, HTML elements nesting, block and inline elements, the head section including title, style, link, meta name, charset, http-equiv, content, script tag, and base tag, character formatting tags such as B, U, I, SUB, SUP, the HR tag, and FONT tag with attributes like SIZE, COLOR, FACE, and alignment. | 6   |
| 2    | Working with Images, Links, and Multimedia: Image element using IMG tag and attributes like SRC, ALT, ALIGN, BORDER, WIDTH, and HEIGHT, introduction to the Anchor tag and HREF attribute for creating hyperlinks, TITLE, NAME, ACCESSKEY, and TARGET attributes, using images as hyperlinks with Anchor tag, use of audio and video tags with attributes.                                                                                                                 | 6   |
| 3    | Tables, Frames, and Forms: Presenting information using TABLE, TR, TD, and TH tags, use of ROWSPAN and COLSPAN, inline frame using IFRAME tag, dividing windows using FRAME and FRAMESET tags, different examples using frameset and inline frames, FORM tag and widgets including INPUT, SELECT, and TEXTAREA, INPUT tag details, Submit and Reset buttons, and form processing with Action and Method attributes.                                                        | 6   |
| 4    | Introduction to CSS with inline styles and the <style> element, basic CSS syntax and selectors, linking external style sheets, creating and managing external style sheets, user style sheets, and understanding conflicting styles. Introduction to JavaScript, versions, variables, let, const, datatypes, operators: Arithmetic, Logical, Comparison, Assignment, concat, Ternary.                                                                                      | 6   |
| 5    | Control Statements: if ... else, switch-case, Loops: for, while, do ... while, break and continue statement, built-in functions-eval, parseInt, parseFloat, user-defined function, alert, prompt, confirm, DOM, document objects: document. write, document.getElementsByTagName                                                                                                                                                                                           | 6   |

#### Suggested Readings:

1. HTML Complete BPB Publication (Sybex)
2. Internet and World Wide Web, H.M. Deitel, P.J. Deitel, A.B. Goldberg, Pearson–Prentice Hall

|                |                |           |    |              |     |
|----------------|----------------|-----------|----|--------------|-----|
| Semester       | BCA I Sem      |           |    |              |     |
| Course Code    | CCBCA122       |           |    | Credits      | 2   |
| Course Name    | WEB DESIGN LAB |           |    | Pds Per Week | 4   |
| Course Type    | Practical      |           |    |              |     |
| MID TERM MARKS | 30             | ESE MARKS | 70 | TOTAL        | 100 |

**Exercises based on:** Basics of HTML & Markup Language Fundamentals: Introduction to HTML, Basic HTML Structure, Character Formatting Tags, Drawing Rules with HR Tag, Anchor Tag A Essentials, Advanced HTML Features & Layout Techniques: Creating Lists & List Items, Styling with Font Tag, Image Integration, Designing Tabular Data & Frame Structures, Enhancing Content Presentation with Semantics & Multimedia: New HTML5 Semantic Tags, Adding Audio & Video Clips, Handling Cascading Style Sheets (CSS), Mastering Web Page Navigation & Interaction: Hyperlinks & Table Elements, Effective Form Usage, Customizable Input Types & Buttons, Scripting Behaviors with JavaScript, Conditionals, Loops, & Document Management: Flow Control Statements: If Else, Switch-Case, Iterative Looping Mechanisms, Built-In JavaScript Functions & Expressions, Accessing DOM Elements with Document Objects.

|                |                                        |           |    |              |     |
|----------------|----------------------------------------|-----------|----|--------------|-----|
| Semester       | BCA I Sem                              |           |    |              |     |
| Course Code    | CCBCA113                               |           |    | Credits      | 4   |
| Course Name    | COMPUTER ORGANIZATION AND ARCHITECTURE |           |    | Pds Per Week | 3   |
| Course Type    | Theory                                 |           |    |              |     |
| MID TERM MARKS | 30                                     | ESE MARKS | 70 | TOTAL        | 100 |

| S. No. | Course Outcomes                                                                                                |
|--------|----------------------------------------------------------------------------------------------------------------|
| CO1    | Understand and apply various number systems, binary arithmetic, and binary codes                               |
| CO2    | Explain and analyse Boolean algebra principles and logic operations                                            |
| CO3    | Design and evaluate logic circuits                                                                             |
| CO4    | Describe the organization of a basic computer system with micro program                                        |
| CO5    | Describe the architecture of a basic computer system and analyse I/O organization and data transfer techniques |

| UNIT | COMPUTER ORGANIZATION AND ARCHITECTURE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | HRS |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1    | Number Systems and Boolean Algebra: Introduction to Number Systems; Decimal, Binary, Octal, Hexadecimal Number Systems; Number System Conversions; Binary Arithmetic; BCD Addition and Subtraction Binary Codes – 8421 Code, ASCII, EBCDIC, Excess-3 code, Gray Code Boolean Algebra – Introduction to Logic; Logic Operations; AND, OR, NOT Laws; Commutative, Associative, Distributive Laws, De-Morgan's Theorems.                                                                                                                      | 6   |
| 2    | Logic Gates and Simplification Techniques: Classification of Gates – Basic Gates (AND, OR, NOT), Universal Gates (NAND, NOR), Exclusive Gates (XOR, XNOR). Boolean Functions – Minterms, Maxterms, SOP, POS. Reduction Techniques – Need for Reduction; Boolean Algebra Simplification; Karnaugh Maps (2, 3 & 4 Variables).                                                                                                                                                                                                                | 6   |
| 3    | Combinational & Sequential Circuits: Combinational Circuits – Half Adder, Full Adder, Subtractor, Decoder, Encoder, Multiplexer Demultiplexer, Magnitude Comparator. Sequential Circuits – Flip-Flops: SR, D, JK, Registers – 4-bit Register with Parallel Load; Bidirectional Shift Register with Parallel Load. Counters – 4-bit Synchronous Counter; 4-bit Asynchronous Counter.                                                                                                                                                        | 6   |
| 4    | Basic Computer Organization & Micro program Control Topics: Basic Computer Organization – Instruction Codes; Address, Data & Control Bus; Computer Registers. Computer Instructions; Timing & Control; Instruction Cycle; Memory-Reference Instructions. Arithmetic & Logic Circuit; Instruction Formats, Addressing Modes. Micro-program Control Organization – Control Memory; Address Sequencing; Mapping of Macro-Operations. Subroutines; Microinstruction Format; Micro-program Example; Micro-program Sequencer; Memory-Mapped I/O. | 6   |
| 5    | Input/output Organization and Data Transfer: Modes of Data Transfer – Asynchronous Data Transfer (Strobe Control, Handshaking), Interrupt-Driven I/O, Direct Memory Access (DMA), DMA Operation, DMA Controller. I/O Organization – I/O Interface, Memory-Mapped I/O vs I/O-Mapped I/O, Programmed I/O.                                                                                                                                                                                                                                    | 6   |

#### Suggested Readings:

1. Digital Logic and Computer Design – M. Morris Mano
2. Computer System Architecture – M. Morris Mano

|                |               |           |    |              |     |
|----------------|---------------|-----------|----|--------------|-----|
| Semester       | BCA I Sem     |           |    |              |     |
| Course Code    | MDBCA111      |           |    | Credits      | 4   |
| Course Name    | MATHEMATICS I |           |    | Pds Per Week | 3   |
| Course Type    | Theory        |           |    |              |     |
| MID TERM MARKS | 30            | ESE MARKS | 70 | TOTAL        | 100 |

| S. No. | Course Outcomes                                                                    |
|--------|------------------------------------------------------------------------------------|
| CO1    | Describe and explain fundamental concepts of sets, functions, and relations.       |
| CO2    | Apply counting techniques and solve recurrence relations in computational problems |
| CO3    | Analyze graph structures and tree properties to solve algorithmic problems.        |
| CO4    | Solve algebraic problems using matrices, determinants, and Eigen concepts.         |
| CO5    | Integrate discrete mathematical concepts to model and solve CS-related problems.   |

| UNIT | MATHEMATICS I                                                                                                                                                                                                                                                                                                                                                                  | HRS |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1    | Set, Set Operations, Properties of Set operations, Subset, Venn Diagrams, Cartesian Products. Functions: Domain and Range, Composition of Functions, Surjective (onto), Injective (one-to-one) and Bijective Functions, Inverse of Functions, Useful Functions for Computer Science: Exponential and Logarithmic Functions, Polynomial Functions, Ceiling and Floor Functions. | 6   |
| 2    | Relations on a Set, Properties of Relations, Representing Relations using Matrices and Digraphs, Types of Relations, Equivalence Relation and Partition on Set, Closures of Relations, Warshall's Algorithm.                                                                                                                                                                   | 6   |
| 3    | Basics of Counting, Pigeonhole Principle, Permutation, Combination, Binomial Coefficients, Binomial Theorem. Recurrence Relations: Modeling Recurrence Relations with Examples like Fibonacci Numbers, Tower of Hanoi Problem. Solving Linear Recurrence Relations with Constant Coefficients using Characteristic Equation Roots Method.                                      | 6   |
| 4    | Basic Terminologies of Graphs, Connected and Disconnected Graphs, Sub graph, Paths and Cycles, Complete Graphs, Digraphs, Weighted Graphs, Euler and Hamiltonian Graphs, Trees: Properties of Trees, Concept of Spanning Tree. Planar Graphs. Definitions and Basic Results on the Topics Mentioned.                                                                           | 6   |
| 5    | Types of Matrices, Algebra of Matrices – Addition, Subtraction, and Multiplication of Matrices Determinant of a Matrix, Symmetric and Skew-Symmetric Matrices, Orthogonal Matrix, Rank of a Matrix, Inverse of a Matrix. Applications of Matrices to Solve System of Linear Equations, Eigen Values and Eigen Vectors, Caley-Hamilton Theorem.                                 | 6   |

**Suggested Readings:**

1. Garg, Reena, Engineering Mathematics, Khanna Book Publishing Company, 2024.
2. Kolman B., Busby R. and Ross S., Discrete Mathematical Structures, 6th Edition, Pearson Education, 2015.
3. Vasishtha A. R. and Vasishtha A. K., Matrices, Krishna Prakashan, 2022.